XIAMEN UNIVERSITY MALAYSIA

Course Code	:	SWE406/DSC301
Course Name	:	Fundamental Research in Academic Project
Lecturer	:	Dr Catherine Hiew Fui Chin
Academic Session	:	2026-04
Assessment Title	:	Proposal
Submission Due Date	:	15th June 2026
Prepared by	:	

Student ID	Student Name
SWE2309520	Li Ruibing

Date Received :

Feedback from Lecturer:	
	Mark:

Own Work Declaration

I/We acknowledge that my/our work will be examined for plagiarism or any other form of academic misconduct, and that the digital copy of the work may be retained for future comparisons.

I/We confirm that all sources cited in this work have been correctly acknowledged and that I/we fully understand the serious consequences of any intentional or unintentional academic misconduct.

In addition, I/we affirm that this submission does not contain any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. This work is my/our original creation, and it has not been based on any work of other students (past or present), nor has it been previously submitted to any other course or institution.

Signature: Li Ruibing 厉蕊冰

Date:2026/6/15

Component Title	Proposal				Percentage (%)	60%
Criteria	**Needs Improvement**	**Satisfactory**	**Good**	**Excellent**	**Mark Allocate**	**Mark Award**
CLO1	**0 - 8**	**9 - 12**	**13 - 15**	**16- 20**	**20**	
1. Introduction (Background), Problem Statement & Objectives	Inadequate background; unclear or trivial problem; missing or weak objectives.	Basic background; somewhat vague or generic problem/objectives.	Relevant background; mostly clear problem and objectives.	Focused and critical background introduction; clear, significant problem identified; specific, measurable, and achievable objectives aligned with the problem.		
CLO1	**0 - 12**	**13 - 18**	**19 - 23**	**24 - 30**	**30**	
2. Literature Review	Inadequate or irrelevant review; no clear synthesis or foundation for the project.	Basic review; limited critical insight; vague or general gaps.	Relevant literature reviewed; some critical insight; research gap generally identified.	Focused and critical review of relevant literature. Critically compares studies with LR metric, integrates findings, identifies gaps, and builds a foundation for a potential solution or theoretical framework justifying the new project.		
CLO2	**0 - 8**	**9 - 12**	**13 - 15**	**16 - 20**	**20**	
3. Methodology & Justification	Inappropriate or missing methodology and research design, no or little justification	Methods or research design are described, but the justification is weak	Appropriate methods and research design with some justification	Research and development methods and research design are clearly explained and justified		
CLO2	**0 - 8**	**9 - 12**	**13 - 15**	**16 - 20**	**10**	
4. Expected Outcomes, Planning & Resources	Weak or missing rationale; unrealistic or unclear outcomes; poor or missing planning.	Limited or unclear rationale; outcomes and planning lack detail.	Adequate rationale and expected outcomes; generally feasible timeline and planning.	Strong and logical rationale linked to literature gap; realistic and meaningful expected outcomes; detailed and feasible Gantt chart and resource planning.		
CLO 3	**0 - 3**	**4 - 5**	**6 - 7**	**8 - 10**	**20**	
5. Ethics, Writing Quality, Structure & Referencing	Ethics is not addressed. Poor writing and structure; major grammar errors; referencing mostly incorrect or missing.	Ethics discussion is weak. Some unclear writing or structural issues; referencing is inconsistent.	Ethics discussion is good. Generally clear and organized; minor writing or formatting issues; mostly correct referencing.	Ethical considerations are fully integrated and appropriate. Professional academic writing; logical structure; highly clear and coherent; free of grammar errors; consistent APA referencing.		
				Total	**100**	

_____ _____

Lecturer: Catherine Hiew Fui Chin **Vetter:**
Date: **Date:**

XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

School of Computing and Data Science (SCDS)
Reference No. : XMUM.SCDS (V6)
Effective Date : 7th October 2025

Undergraduate FYP Proposal Form[1]

Section I: Student's Information:

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Section II : Industry Information: (Complete this section if your project is going to be conducted in partnership with any company)

Company Name			
Contact Person			
Designation		**Phone**	
Email ID			

Section III : Declaration by the Student

By signing this form, I confirm that I have read and will adhere to the **Final Year Project Proposal Guidelines** of Xiamen University Malaysia as applicable to this application.

Signature of Student: Li Ruibing 厉蕊冰 Date: 13th June 12, 2026

Section VI : Supervisor Approval

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[1] The length of the proposal should be between 2000-3000 (Excluding proposed title & references)

SECTION V : Proposal Information

Proposed Title *(Capitalise each word)* ***(Not more than 20 words)***

A Hybrid RAG-FSM Architecture for Internet-Delivered Cognitive Behavioral Therapy for Insomnia

Introduction *(Introduce the context & research gaps/problem)* ***(300-400 words)***

Insomnia disorder, also known as chronic insomnia, is a serious public health problem. International clinical guidelines recommend Cognitive Behavioral Therapy for Insomnia (CBT-I) as a first-line nonpharmacological intervention (Morin et al., 2006; Riemann et al., 2017). CBT-I contains a series of structured behavioral therapies, including sleep restriction, stimulus control, and sleep hygiene education, alongside cognitive restructuring targeting negative and dysfunctional sleep beliefs. A large amount of evidence shows that CBT-I can continuously improve sleep efficiency and insomnia severity (Ritterband et al., 2017).

Although the effect of face-to-face CBT-I has been confirmed, it is difficult to promote this therapy because there are not enough therapists, too many people waiting in line, the high cost of treatment and geographical restrictions (Espie et al., 2012). In response to this problem, internet-delivered CBT-I (iCBT-I) has been confirmed to be a viable alternative. It is scalable, effective and relatively low-cost (Ritterband et al., 2017; Zachariae et al., 2016). However, the current iCBT-I systems still face many obstacles, such as high patient attrition, poor treatment compliance, and a lack of strong natural language capabilities to provide personalized interaction (Koffel et al., 2016; Lin et al., 2025).

Large Language Models (LLMs) can generate empathetic dialogues and understand natural language, which bring new possibilities to the field of digital mental health, help to improve user participation, and can also provide personalized treatment content (Bao et al., 2025). However, directly deploying LLMs in clinical environment will also bring some serious risks, such as inaccurate suggestions, content hallucination and possible deviations from standard treatment process (Bedi et al., 2025; Mansoor, 2025). As for the existing LLM fine-tuning methods for CBT-I, they also face two main problems. The ability of clinical reasoning will decrease during long-term treatment, and the calculation expense is too high. (Chen et al., 2024).

One problem that this project wants to solve is how to balance the flexibility of LLM in dialogue with the clinical rigor of the structured treatment process and protocols. CBT-I is actually well suitable for implementing in a structured computation method, because its behavioral part runs step by step in the set time sequence according to clear rules. Yet the cognitive reconstruction in CBT-I requires a more flexible and situation-based response to those sudden dysfunctional ideas. This project proposes a hierarchical structure. This architecture combines LangGraph-based finite state machine (FSM) control to manage the behavioral stage, and also integrates retrieval enhancement

generation (RAG) technology to achieve clinically compliant cognitive intervention. This unified dialog framework can not only ensure that the standardized clinical protocols are strictly followed, but also respond in an adaptive manner.

Aims and Objectives *(What & how)* ***(100-150 words)***

The aim of this project is to design and develop an iCBT-I dialog system prototype, which integrates RAG technology and FSM workflow orchestration. The proposed architecture is expected to combine the flexibility of LLM and deterministic process control to support clinically compliant insomnia interventions. During the whole implementation of CBT-I, the system is able to improve the consistency of workflow, enhance the reliability of knowledge retrieval and protocol adherence.

Objectives:

- To develop an iCBT-I prototype, which integrates FSM workflow and RAG knowledge retrieval.
- To implement structured CBT-I intervention process based on the proposed architecture.
- To evaluate the prototype functionality through scenario-based testing and simulated dialogues.
- To validate the operating efficiency of the iCBT-I prototype using quantitative metrics.

Literature Review *(Related works)* ***(1000-1300 words)***

3.1. Clinical Foundations of CBT-I

CBT-I is a multicomponent psychological intervention method that integrates several behavioral strategies, including sleep restriction therapy (SRT), stimulation control therapy (SCT) and sleep hygiene education, as well as cognitive reconstruction techniques for those beliefs related to poor sleep (Morin et al., 2006). The American Academy of Sleep Medicine commissioned a comprehensive meta-analysis, which confirmed that CBT-I can stably improve many key sleep indicators, so it is recommended as an effective treatment for chronic insomnia (Morin et al., 2006). Epstein et al. (2012) further verified through a dismantling randomized controlled trial (RCT) that although a single behavioral component itself can also bring a certain improvement effect, combining multiple components for intervention can achieve the highest remission rate**,** and is desirable in clinical practice.

3.2 Evolution of Internet-Delivered CBT-I

The shift from face-to-face treatment to iCBT-I is promoted by step-by-step empirical validation. Ritterband et al. (2009) established the first confirmed the feasibility of fully automatic digital treatment that does not require the participation of therapists through the SHUTi project, a six-week web-based CBT-I project. Later, a large-scale RCT verified that this treatment effect continued in the follow-up visit after one year, which made SHUTi a benchmark for measuring the efficacy of iCBT-I (Ritterband et al., 2017). But at the same time, there are still problems of low treatment compliance and high patient dropout rate in digital treatment forms (Lin et al., 2025). Another mobile phone application named CBT-i Coach, which is a rule-based expert system, uses hard-coded

algorithms to calculate sleep prescriptions. A pilot trial of it showed that structured digital tools can well improve the retention rate of treatment (Koffel et al., 2016). However, as a purely deterministic system, CBT-i Coach lacks the ability to understand natural language and cannot carry out cognitive reconstruction. Because of this functional defect, people began to promote the integration of conversational artificial intelligence (AI) into the treatment of iCBT-I. Recently, Chiu et al. (2025) conducted an RCT to compare a CBT-I chatbot and a sleep education website to see how effective they are in psychiatric outpatients. The test results showed that the sleep quality and comprehensive psychological symptoms of the patients in the group of chatbots were highly improved. This also provided direct clinical evidence for chatbot-based CBT-I treatment, indicating its feasibility.

3.3 LLMs in Mental Health: Promise and Pitfalls

The emergence of Large Language Models (LLMs) has brought new possibilities to AI-driven mental health interventions, especially in the realization of natural language interaction and personalized dialogue generation. Inkster et al. (2018) provided some early real-world evidence that an empathy-driven conversational AI application called Wysa can well alleviate depressive symptoms, which also suggested that users are willing to accept text-based AI treatment support. Another example is Limbic Care, after large-scale RCT evaluation, it was found that compared with the form of static content, interactive CBT supported by LLM can greatly improve user participation, and personalized functions can also bring better symptom relief. (McFadyen et al., 2024). Gupta et al. (2024) showed that integrating Retrieval-Augmented Generation (RAG) into mental health chatbots can improve the reliability of high-risk intent classification, validating retrieval-based enhancement as a safety mechanism for clinical AI applications.

However, the critical limitations still exist. Bedi et al. (2025) systematically evaluated LLMs for CBT dialogue in both generation-only and RAG-enhanced settings. They found that while LLMs produce linguistically adequate therapeutic responses, they remain limited in maintaining treatment consistency and conveying empathy across sessions, indicating that the quality of language itself does not ensure the effectiveness of treatment. Chen et al. (2024) fine-tuned ChatGLM-6B with LoRA for a mobile CBT-I application, and reported encouraging initial outcomes, but they also pointed that parameter-efficient fine-tuning is prone to reasoning degradation in long-term multi-session treatment interactions. Most comprehensively, Bao et al. (2025) evaluated eight LLMs with three adaptive strategies on an RAG-enhanced iCBT-I dialogue system, and concluded that clinical reliability demands more than prompt engineering or model fine-tuning; additional architectural constraints are essential.

3.4 FSM and RAG: Architectural Solutions

Finite State Machines (FSMs) provide a well-established formalism for modelling systems with discrete states and deterministic transitions, making them well-suited for clinical workflow management where protocol compliance is essential. Recent work has begun to explore FSM-based

control within AI-driven therapeutic dialogue. Zhao et al. (2025) proposed FiSMiness, a paradigm that applies FSM to structure LLM-based emotional support conversations by defining states as the combination of user emotion and supporter strategy, demonstrating that FSM-guided generation outperforms both direct inference and chain-of-thought prompting on established benchmarks. Vijayan (2025) implemented a hybrid dialogue system, which integrates BiLSTM intent classification with FSM-controlled dialogue flow, specifically for structured health counselling. This work found that the FSM framework can be used in medical applications while taking into account both linguistic flexibility and operational reliability. In deployed clinical environments, the CLARITY triage system also confirmed that it is feasible to combine FSM dialogue management and LLM analysis on a large scale (CLARITY, 2025). LangGraph is a state machine framework based on graph structure, which is used to arrange LLM. It supports conditional routing, state checkpoints and interruptions of Human-Computer Interaction (HCI), which directly fit the requirements of CBT-I in behavioral stage management (LangChain, 2024).

Retrieval-Augmented Generation (RAG) method proposed by Lewis et al. (2020) solves a fundamental limitation of parametric LLM. RAG links the generated output with the retrieved external knowledge, which can reduce the subjective assumption of the model and improve the accuracy of the facts. In the field of mental health, RAG has been used to link LLM's answers with clinical guidelines and diagnostic standards in psychiatric diagnostic tasks (Mansoor, 2025). It can also support culturally adaptable CBT dialogue by retrieving information from the knowledge base (KB) of specific situations (Forane et al., 2025). Skill-RAG indicated that the model can use its own knowledge to filter out irrelevant retrieval contents, which is used to judge whether it can ensure that the cognitive intervention of CBT-I always maintains clinical applicability and situational relevance (Isoda, 2025).

Rather than separately deploying LangGraph or KBs, Lenh et al. (2024) showed that FSM and RAG can be integrated into the same mental health dialog system. In this system, a module controlled by FSM is responsible for managing depression assessment, and another RAG-based subsystem is responsible for handling open mental health queries, and the two can be flexibly switched through an intention detector. But this integration method is still limited to general depression support, and it does not solve several key problems required by CBT-I, such as structured time-dependence, deterministic sleep prescription calculation, or the global interruptible cognitive reconstruction.

Discussion and Summary

The rule-based systems can ensure clinical compliance of certainty, but they lack the ability to interact with natural language. The LLM-based system can improve the user's participation and dialogue flexibility, yet it cannot guarantee the strict adherence of the structured treatment plan. While the enhanced system based on RAG can improve the factual basis of the answer, the inherent time dependence in behavioral sleep intervention cannot be captured. Recent hybrid systems have

started integrating FSM with LLM or RAG components (Zhao et al., 2025; Lenh et al., 2024), but the problem is that these architectures just use flat state machines to manage relatively single tasks, such as questionnaires or emotional support. They cannot handle the multi-stage and time-dependent behavioral progress required in CBT-I. No current system has been able to combine hierarchical FSM-driven behavioral workflow orchestration with cognitive intervention, which is based on RAG separately tracking the behavioral layer and the cognitive layer in a framework specially designed for CBT-I. As shown in Figure 3.1, this project proposes a hierarchical RAG-FSM architecture. In this architecture, a multi-layered FSM is responsible for deterministic behavioral stage progression, while a cognition-focused RAG module runs as a global interrupt mechanism.

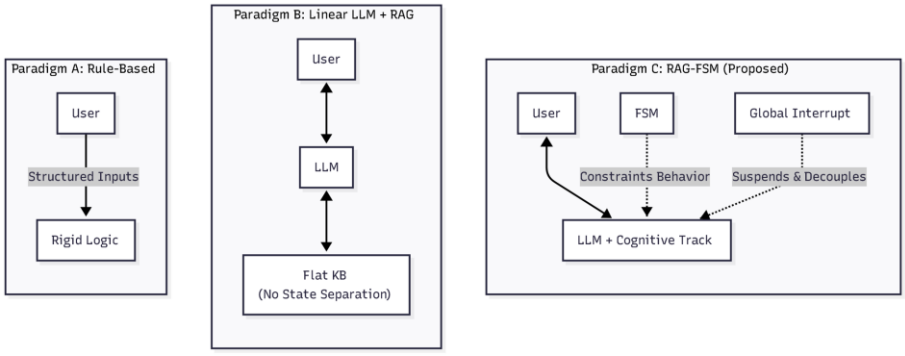


Figure 3.1 Conceptual comparison of rule-based, linear LLM-RAG, and the proposed FSM-RAG conversational architectures

Research Methodology

(Data collection / system design, tools & technologies / datasets, & Evaluation methods (Ex. testing, user study, simulation) ***(300-500 words)***

4.1 Prototype Development Method

This project uses the Design Science Research method. The whole process is to repeatedly build, demonstrate and evaluate the workpiece through an iterative cycle. An Agile SDLC method is used for system development to combine system implementation with iterative evaluation. Figure 4.1 shows the whole method framework and its corresponding stages.

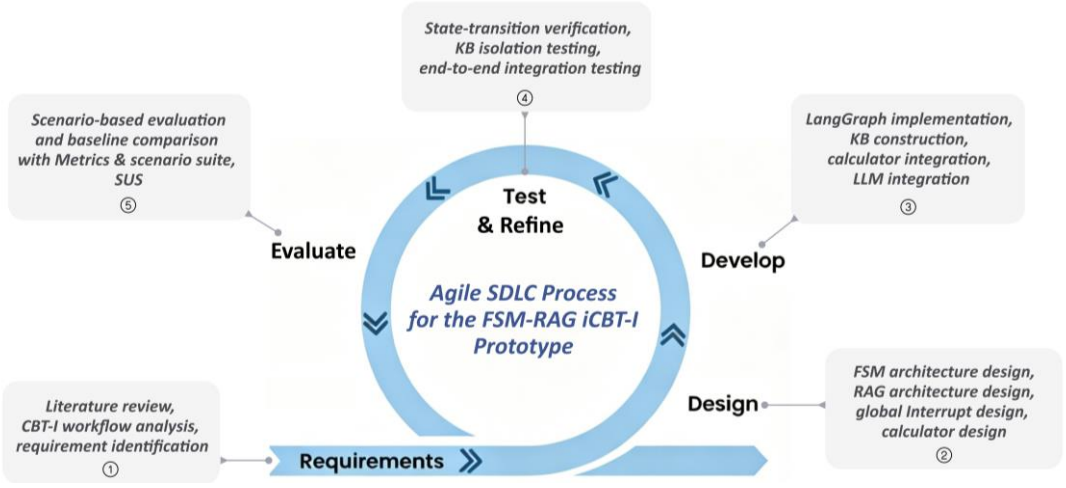


Figure 4.1 Stages of Agile SDLC and Corresponding Activities

4.2 System Architecture Design and Development

The prototype implements a hierarchical architecture containing four integrated components.

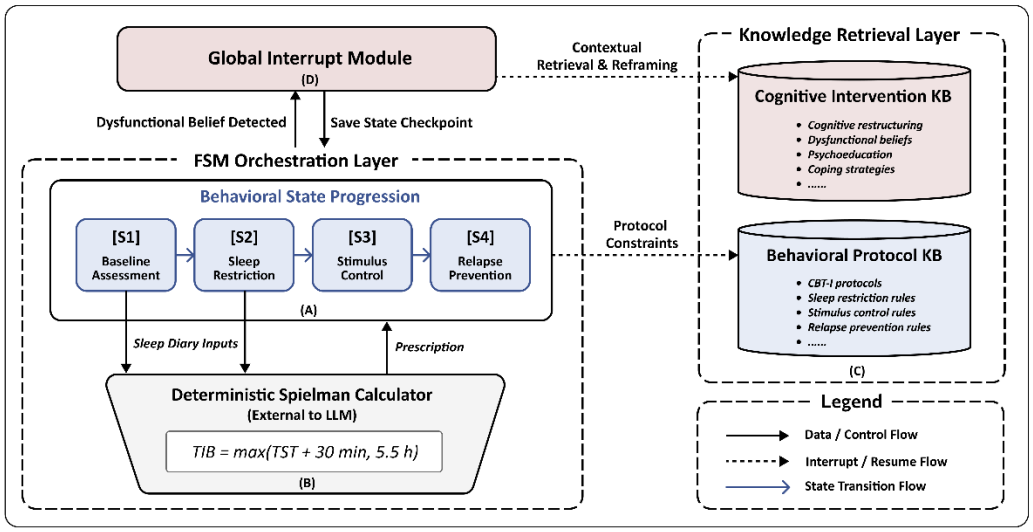


Figure 4.2 Hierarchical Architecture of the RAG-FSM iCBT-I System

A. The FSM Orchestration Layer uses LangGraph, which manages the four sequential execution and deterministic CBT-I stages ($\boldsymbol{S_1}$-$\boldsymbol{S_4}$) according to the established protocol.

B. In $\boldsymbol{S_1}$-$\boldsymbol{S_2}$, the system collects the sleep diary inputs through structured dialogue prompts, which then be sent to a deterministic Spielman calculator independent of LLM.

C. The Knowledge Retrieval Layer uses RAG, which is composed of two isolated vector KBs, Behavioral Protocol KB and Cognitive Intervention KB.

D. The Cognitive Interrupt Module detects the dysfunctional belief, it first suspends the currently running behavioral state, and call the cognitive intervention KB via LangGraph checkpoint, then resumes it back after processing.

4.4 Testing and Refinement

This stage needs to iteratively verify the integrated system to ensure the system quality before the formal evaluation begins. The continuous testing work is divided into three parts:

(1) Check the state conversion during the execution process to validate the FSM logic.

(2) Perform boundary leakage testing to guarantee strict isolation between the dual KB.

(3) Conduct end-to-end integration testing to validate the fluid data flow, Spielman calculation synchronization, the RAG pipelines, and the LLM routing layer.

Prior to evaluation, the expected system behavior is specified for every test case, drawing from the CBT-I clinical protocol for FSM transition logic, the Spielman algorithm for prescription benchmarks, and published CBT-I therapist manuals for interrupt trigger.

4.5 Evaluation

(1) Architectural Metrics Selection

Table 4.5.1 presents four evaluation metrics, each mapped to one architectural component and one documented failure mode.

No.	Metric	Definition	Target	Architectural component	Failure mode addressed
(a)	FSM Transition Accuracy	Proportion of correct stage transitions	$\geq 95\%$	FSM Orchestration Layer	Conversational Protocol Drift (Bao et al., 2025)
(b)	Prescription Precision	Accuracy of the Spielman formula output	100%	Deterministic Calculator	LLM clinical arithmetic failure (Chen et al., 2024; Bao et al., 2025)
(c)	Interrupt Precision and Recall	Precision and recall of interrupt trigger decisions	$P \geq 80\%$, $R \geq 75\%$	Cognitive Interrupt Module	Unreliable trigger decisions (Gupta et al., 2024)
(d)	Contextual Recoverability	Correct recovery rate after each cognitive interrupt	$\geq 90\%$	FSM-Interrupt Interface	Inter-session context loss (Bedi et al., 2025)

Table 4.5.1 Evaluation Metrics and Validation Targets

(2) Scenario-Based Test Suite Preparation

Six scenario categories function as test environments targeting the four metrics in Table 4.5.1. Scripted dialogues are applied in each category. Tests cover functional integration, state transition and retrieval-isolation integrity.

#	Category	Sample Case	Property Tested	Metric
1	Workflow progression	Compliant 7-day diary *"Bedtime: 11 PM; Wake-up: 6 AM; Awakenings: 0. SE is above 85% this week"*	FSM orchestration under compliant path	(a)
2	Workflow deviation	Off-topic tangents *"What is the weather today?"*	Resistance to protocol drift	(a)
3	Contradictory input	Logically conflicting values *"I woke up 15 times last night but my sleep efficiency was 98%."*	FSM verification and anomaly flagging	(a) (b)
4	Interrupt trigger	Clear dysfunctional statements *"If I don't fall asleep in 10 min, my whole week will be ruined."*	Cognitive interrupt triggering (Recall)	(c)
5	False-positive suppression	Standard complaints or negative mood text *"I feel extremely exhausted and irritable today due to poor sleep."*	Workflow continuity preservation (Precision)	(c)
6	Post-interrupt recovery	[Completed cognitive restructuring] *Let's return to your sleep window...*	Checkpoint-based context restoration	(d)

Table 4.5.2 Scenario Test Suite

(3) Baseline Simulation

A single-factor controlled experiment quantifies the FSM's architectural contribution. The independent variable is the presence or absence of LangGraph StateGraph routing; dependent variables are the four architectural metrics; control variables are prompt templates, model weights, and RAG content. The baseline runs the same LLM and RAG assets as a linear chain without FSM state management.

(4) Supplementary Usability Assessment

The usability assessment targets the software interface, not the therapeutic intervention. Peer usability testing recruits N = 5–10 volunteers via convenience sampling. Participants complete the System Usability Scale (SUS) after scripted sessions.

4.6 Tools, Technologies, and Data Sources

- LangGraph
- Python development environment
- Open-source LLM
- Vector database (e.g., FAISS)
- Scenario-based test scripts
- CBT-I clinical guidelines and intervention manuals
- System Usability Scale questionnaire

Scope & Limitations *(100-150 words)*

The scope of this project is limited to the design and evaluation of an iCBT-I prototype based on FSM-RAG. This system only focuses on selected CBT-I modules containing sleep diary management, sleep restriction guidance, and cognitive restructuring support. The evaluation is for the performance metrics of the system itself and preliminary usability assessment, but its clinical effect or treatment results are not validated. As a prototype for research, it is not suitable for deployment in a formal clinical environment, and cannot replace professional medical services. A limited number of scripted scenarios are used in the evaluation process and a small convenience sample for usability testing, which will reduce the universality of the results a certain extent.

Potential Project Significance *(100-200 words)*

From the theoretical view, the FSM-RAG architecture provides a model that can be reused. This model does not need to fine-tuning LLM, it can be integrated into the structured workflow in clinical practice, and meanwhile, it can ensure the safety of patients through the execution of certain constraints. It balances the reliability of rule-based systems and the flexibility of conversational AI, puts forward some design principles that can be migrated, and sets a scenario-based evaluation

framework for clinical AI architecture. From the practical view, this architecture can help build a stable and compliant iCBT-I system. The decision-making boundaries of this system are ethically transparent, which helps to overcome some limitations of existing online intervention tools. It improves digital insomnia intervention through local deployment models, so as to prevent sensitive health data from being transmitted to the outside, and this architecture can also be extended to other structured digital psychotherapy applications.

Expected Outcomes & Concluding Remarks

(Deliverables, expected results & conclusions) ***(200-300 words)***

7.1 Expected Outcomes

The tangible deliverable is a functional iCBT-I prototype application featuring:

1) A LangGraph-powered conversational interface that manages behavioral therapy progression through a Macro-FSM with checkpointed state persistence.
2) A dual-track RAG knowledge base serving both behavioral protocol retrieval and cognitive intervention retrieval.
3) An integrated sleep diary module with automated sleep prescription calculation following the clinically validated Spielman formula.
4) A cognitive interrupt mechanism that dynamically invokes targeted cognitive restructuring when dysfunctional sleep beliefs are detected.

Measurable outcomes and corresponding performance thresholds are listed in Table 4.2.1. Besides, a SUS score above 68 is targeted to indicate acceptable usability.

7.2 Conclusion

Solve the delivery problem of iCBT-I from the perspective of software architecture rather than simple model capability, this project contributes a novel and potentially generalizable design pattern for conversational AI that meets clinical standards. The hierarchical FSM-RAG architecture responds to the demand identified across the literature, that is, the need for an architectural constraint mechanism that can enable LLMs to operate safely in the evidence-based therapy framework. The evaluation strategy in the plan combines architecture performance metrics with clinical outcome measures, which can provide multidimensional evidence for the feasibility of the method. The successful demonstration of the prototype system is expected to provide a reference for the future supervised AI-based digital therapy design and support the responsible integration of generative AI into the mental health delivery systems.

Key References

(Recent references APA 7th Edition) ***(10-20 references APA style)***

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Gantt Chart

Task	October					November				December			
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
Phase 1: Requirements & Design													
Software Requirements Specification													
System Architecture Design													
Evaluation Framework Design													
Phase 2: Prototype Development													
Environment Setup													
Core System Implementation													
Knowledge Base Construction													
System Integration & Debugging													
Phase 3: Testing & Refinement													
Gold-Standard Reference Construction													
Verification & Integration Testing													
Phase 4: Evaluation													
Scenario Testing & Baseline Comparison													
Usability Assessment (SUS)													
Data Analysis & Results													
Phase 5: Thesis Writing													
Initial Thesis Drafting													
Methodology & Implementation Documentation													
Revise and Conclude													

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[2] Please follow APA style of referencing.

Appendix 1: Key Literature Review Synthesis Matrix

Source	Concept 1: Digital CBT-I	Concept 2: LLM in Mental Health	Concept 3: FSM / Workflow Control	Concept 4: RAG / Knowledge Base	Gap & Limitations	Potential Innovation
Morin et al. (2006)	First-line CBT-I efficacy benchmark via meta-analysis	—	—	—	Pre-digital; no AI consideration	—
Epstein et al. (2012)	Multicomponent CBT-I outperforms single components in remission	—	—	—	Supports integrated delivery but no digital implementation	—
Ritterband et al. (2009; 2017)	SHUTi: first automated web CBT-I; sustained efficacy at 1-year follow-up	—	—	—	Non-conversational; static content; no NLP	Apply NLP/LLM for conversational delivery
Koffel et al. (2016)	Rule-based CBT-i Coach app reduced dropout to 0%	—	Implicit FSM via hard-coded branching	—	No NLU or cognitive restructuring capability	Apply LLM for NLU; use as baseline to compare
Chiu et al. (2025)	RCT: chatbot CBT-I improved sleep quality and comorbid symptoms	Direct clinical evidence for chatbot-delivered CBT-I	—	—	Proven chatbot efficacy but no FSM/RAG architecture	Add FSM+RAG architecture on top of proven chatbot model
Inkster et al. (2018)	—	Wysa: empathy-driven AI chatbot; established user acceptability	—	—	Not CBT-I specific; pre-RAG; limited clinical rigor	Design CBT-I-specific dual-track KB
McFadyen et al. (2024)	—	Limbic Care: GenAI tripled user engagement over static workbook	—	—	Engagement over compliance; no workflow control or sleep calculation	Integrate FSM for protocol compliance; add deterministic calculation
Gupta et al. (2024)	—	RAG improved safety-critical intent detection in Wysa	—	RAG as safety guardrail for high-risk intents	Black-box prompt gating; no FSM; not CBT-I specific	Integrate FSM to control CBT-I workflow
Bedi et al. (2025)	—	LLMs produce CBT-like text but fail in empathy and consistency	—	RAG improves grounding but not therapeutic fidelity	Language quality alone does not ensure therapeutic fidelity	FSM enforces cross-session consistency
Chen et al. (2024)	Mobile CBT-I app with LoRA fine-tuned ChatGLM-6B	LoRA personalizes dialogue but degrades reasoning over extended use	—	—	Fine-tuning prone to catastrophic forgetting in long-term treatment	Use RAG over fine-tuning for flexibility and stability
Bao et al. (2025)	RAG-enhanced eCBT-I dialogue with structured KG	Benchmarked 8 LLMs with 3 adaptation strategies; fine-tuning alone insufficient	—	RAG+KG enhances clinical response grounding	No FSM; explicitly calls for architectural constraints beyond fine-tuning	Integrate FSM as the architectural constraint layer

Zhao et al. (2026)	—	FSM structures LLM emotional support dialogue	State: utterance, emotion, strategy; outperforms CoT and SFT	—	Validated for general ESC; flat FSM; not CBT-I specific	Hierarchical FSM + dual-track KB for CBT-I
Vijayan (2025)	—	—	BiLSTM+FSM hybrid for health counseling; bridges flexibility and reliability	—	Health-domain FSM validated; no RAG or CBT-I temporal modeling	Add RAG KB and CBT-I temporal stage management
Lenh et al. (2024)	—	Hybrid system for student mental health	FSM controls structured depression questionnaire flow	RAG-based open-ended mental health QA	FSM+RAG co-exist but flat FSM for single task; no dual-track KB	Hierarchical FSM + isolated behavioral KB and cognitive KB
CLARITY (2025)	—	—	FSM+LLM clinical triage at scale; surpassed human routing precision	—	Triage-only; no RAG KB or therapeutic intervention	Extend FSM+LLM to full therapeutic workflow with RAG
Lewis et al. (2020)	—	—	—	Original RAG: parametric + non-parametric memory reduces hallucination	General NLP; clinical use needs curated domain KB	Dual-track KB: behavioral protocol + cognitive intervention
Mansoor (2025)	Scoping review: 4 LLM use cases for sleep	Evidence preliminary; limited external validity	—	RAG as emerging personalization strategy	No FSM+RAG integration found in sleep/CBT-I literature	Integrate FSM+RAG for CBT-I

Appendix 2: Behavioral Protocol KB Examples

Behavioral Protocol KB – Data Sources & Knowledge Base Population Reference

#	Source	Content Summary	CBT-I Stage	Knowledge Topic	Data Type	Items
1	*Spielman et al. (1987)*	Original SRT protocol: TIB = avg TST; min TIB floor = 4.5–5.5h; weekly SE review; SE>=85% → increase, SE< 80% → decrease	S2: Sleep Restriction	Sleep Restriction Algorithm	Clinical algorithm	~8 rules
2	*Morin et al. (2006)*	Meta-analysis of 37 RCTs (N=2,246); 5 empirically supported CBT-I components	S1–S4 (all)	CBT-I Component Efficacy	Meta-analysis	37 studies
3	*Manber & Carney (2015)*	Session-by-session CBT-I manual: case formulation, SRT decision trees, SCT rules, sleep hygiene scripts, relapse prevention checklist	S1–S4 (all)	Therapist Manual/Decision Trees	Clinical manual	~30 rules
4	*Perlis et al. (2005)*	8-session CBT-I structure: assessment + hygiene → SRT + SCT → cognitive restructuring → relapse prevention	S1–S4 (all)	Session Structure & Homework	Clinical manual	~40 steps
5	*Edinger & Means (2005)*	SRT step-by-step: initial TIB = mean TST from 2-week diary; adjust 15 min/week; alternative SE thresholds (90%/85%)	S2–S3	SRT & SCT Operationalization	Review / Decision rules	~15 rules
6	*Riemann et al. (2017)*	European guideline: CBT-I as first-line (Grade A); 4–8 sessions; SRT+SCT as core; sleep hygiene adjunct only	S1–S4 (all)	Clinical Practice Guideline	Guideline	~20 rec.
7	*Carney et al. (2012)*	Consensus Sleep Diary: 9 core fields (bedtime, SOL, WASO, NWAK, etc.); TST=TIB−SOL−WASO; SE=TST/TIB × 100%	S1→S2	Sleep Diary Data Schema	Standardized instrument	9+3 fields
8	*Drummond & Bei (2024)*	Two-stage SRT: Build (TIB never increases) → Maintain (gradual increase); 83.9% transition within 3 weeks	S2	SRT Stage Model	Empirical study	~6 rules

9	*Epstein et al. (2012)*	Dismantling RCT: combined SRT+SCT highest remission; older adults need modified SE thresholds	S2–S3	Component Efficacy	RCT	N=179
10	*Koffel et al. (2016)*	CBT-I Coach app: sleep diary module, automated prescription, implicit FSM via branching	S1–S4 (ref.)	Rule-Based System Baseline	Pilot RCT	~12 states
11	*Ritterband et al. (2017)*	SHUTi: 6-week automated eCBT-I; 56.6% remission at 1 year; d=2.32; NNT=3.3	S1–S4 (ref.)	Automated eCBT-I Baseline	RCT	N=303
12	*Morgenthaler et al. (2006)*	AASM parameters: SRT+SCT standard (Grade A); cognitive restructuring Grade B; min 4 weeks	S1–S4 (all)	AASM Practice Parameters	Guideline	~12 params
13	*Glovinsky & Spielman (1991)*	Refined SRT: TIB=TST+30min variant; older adult modifications (80% SE threshold); edge-case handling	S2	SRT Protocol Variants	Case studies	~10 cases
14	*Bootzin & Epstein (2011)*	SCT 6 rules: bed only for sleep; leave bed if awake >20min; return when sleepy; fixed wake; no napping	S3	Stimulus Control Rules	Review	6 rules

Appendix 3: Cognitive Protocol KB Examples

Cognitive Intervention KB — Data Sources & Dysfunctional Sleep Belief Taxonomy

#	Source	DBAS Factor	Distortion Sub-Type	Example Dysfunctional Belief	Evidence-Based Reframing
1	Morin et al. (2007)	F1: Consequences	Catastrophic daytime impairment	"If I do not sleep 8 hours, I will not function at work tomorrow."	Sleep need is individual; many function well on 6–7h. One poor night does not eliminate coping ability.
2	Morin et al. (2007)	F1: Consequences	All-or-nothing thinking	"My whole week is ruined because I slept poorly last night."	Sleep quality varies naturally. One poor night does not determine the week. Performance recovers quickly.
3	Morin et al. (2007)	F1: Consequences	Health catastrophe	"If I do not sleep, I will get seriously ill or have a heart attack."	Occasional poor sleep does not cause acute illness. Body has homeostatic mechanisms. Focus on controllable factors.
4	Morin et al. (2007)	F2: Worry/Helplessness	Perceived loss of control	"I have completely lost control of my sleep; nothing will fix it."	CBT-I has strong evidence: 56.6% remission. Sleep is physiological; behavioral strategies can improve it.
5	Morin et al. (2007)	F2: Worry/Helplessness	Anticipatory bedtime anxiety	"I am terrified of going to bed because I know I will lie awake."	Conditioned arousal is common. SRT builds sleep pressure naturally. If not asleep in ~20min, leave bed.
6	Morin et al. (2007)	F2: Worry/Helplessness	Hopelessness about treatment	"I have tried everything. This app probably will not help either."	CBT-I uses behavioral strategies with empirical support, different from general tips. Let us gather diary data first.
7	Morin et al. (2007)	F3: Sleep Expectations	Rigid duration requirement	"I absolutely need 8 hours of perfect uninterrupted sleep."	Sleep need varies (7–9h for adults). Some function well on 6–7h. Use diary to find your optimal window.
8	Morin et al. (2007)	F3: Sleep Expectations	Perfectionism about continuity	"If I wake up even once, my sleep quality is zero."	Brief awakenings are normal (5–15 per night). Total sleep time matters more than continuity.
9	Morin et al. (2007)	F3: Sleep Expectations	Age-inappropriate expectation	"At 65+, I should still sleep 8 hours like at 30."	Sleep architecture changes with age. 6–7h is common and healthy for older adults. Goal is daytime restedness.
10	Morin et al. (2007)	F4: Medication Beliefs	Medication inevitability	"Sleeping pills are the only way I can fall asleep."	CBT-I is first-line treatment. Behavioral strategies address underlying factors. Discuss med changes with doctor.
11	Morin et al. (2007)	F4: Medication Beliefs	Fear of dependence	"I am terrified of becoming addicted but cannot sleep without them."	CBT-I can gradually reduce reliance via SRT/SCT. Must coordinate with prescribing physician.
12	Harvey (2002)	Selective Attention	Clock-watching	"I check the clock 10+ times. At 3 AM, I panic calculating hours left."	Clock-watching increases arousal. Turn clock away. If awake ~20min, leave bed. Trust sleep drive.

13	Harvey (2002)	Selective Attention	Body sensation scanning	"I feel my heart racing; this means I will never fall asleep."	Heightened body awareness is anxious arousal, not impossibility. Try body-scan relaxation or quiet activity.
14	Harvey (2002)	Distorted Perception	Sleep state misperception	"I did not sleep a single minute all week." (diary shows SE 85%+)	Sleep state misperception is real experience. Diary data as collaborative evidence, not to dismiss.
15	Harvey (2002)	Safety Behaviors	Avoidance behavior	"I must cancel my meeting because I might not sleep tonight."	Cancelling reinforces fragility belief. Engaging despite poor sleep builds confidence and reduces threat.
16	Espie et al. (2006)	AIE Pathway	Sleep effort / paradox	"I try SO hard to sleep but the harder I try the more awake I feel."	Sleep is automatic; effort disrupts it. SRT/SCT reduce opportunity to try. Paradoxical intention can help.
17	Carney & Edinger (2006)	Critical Beliefs	Top discriminating belief	"I am worried I have lost control over my sleep." (largest effect size)	CBT-I restores agency through behavioral strategies. Weekly diary data demonstrates progress.
18	Edinger & Carney (2015)	Cognitive Restructuring	Guided exercise template	Identify belief → Rate 0–100% → Examine evidence → Generate balanced thought → Re-rate	Structured Socratic questioning framework from validated CBT-I workbook.

Appendix 4: Evaluation Dialogue Set Examples

Interrupt Trigger Evaluation Set - Scenario Test Suite Examples

ID	Category	Type	User Input	FSM State	Expected System Behavior	Interrupt?
WP-01	1. Workflow	**Compliant diary**	Last night I went to bed at 11 PM, took 20 min to fall asleep, woke once for 10 min, out of bed at 6:30 AM.	**S1 (Day 3)**	Extract diary fields. Compute TST=7h, TIB=7.5h, SE=93.3%. Store. Prompt next day.	**N**
WP-02	1. Workflow	**S1–S2 transition**	(Day 7) Bedtime: 10:30 PM, asleep in 30 min, woke twice totaling 45 min, up at 6:00 AM.	**S1 (Day 7)**	7-day avg SE=77.3%. Trigger S1–S2. Prescribe TIB=6.3h sleep window.	**N**
WP-04	1. Workflow	**S2–S3 transition**	My SE has been above 85% for two weeks and the sleep window is working well.	**S2**	Verify SE≥85% ×2 weeks. Trigger S2–S3. Introduce SCT rules.	**N**
WD-01	2. Deviation	**Off-topic**	What is the weather like today? I want to know if I should go for a walk.	**S2**	Detect off-topic. Acknowledge + redirect to sleep diary. Remain S2.	**N**
WD-03	2. Deviation	**Privacy concern**	Are you using ChatGPT? Is my data being sent somewhere?	**S1**	Transparency response. System runs locally. Data stays on device. Continue.	**N**
WD-05	2. Deviation	**Emotional venting**	I had a terrible fight with my partner. I do not even want to think about sleep.	**S2**	Empathy + offer pause or brief check-in. Remain S2 or pause.	**N**
CI-01	3. Contradiction	**Impossible values**	I woke up 15 times but my SE was 98% and I slept 8 hours.	**S2**	Flag contradiction. Prompt for clarification on awakening duration.	**N**
CI-03	3. Contradiction	**SE boundary**	SE = exactly 85.0%	**S2**	SE at threshold. Maintain if 1st week; increase if 2+ consecutive weeks.	**N**
IT-01	4. Interrupt	**Belief+ (F1)**	If I do not fall asleep in 10 min, my entire week will be ruined. I cannot function.	**S2**	Detect F1. Trigger interrupt. Retrieve reframing. Deliver + resume S2.	**Y**
IT-02	4. Interrupt	**Belief+ (F2)**	I have completely lost control. Nothing makes any difference. I am doomed.	**S1**	Detect F2 helplessness. Trigger interrupt. Deliver reframing. Resume S1.	**Y**
IT-03	4. Interrupt	**Belief+ (F3)**	I need exactly 8 hours unbroken. Anything less means I failed. I could do it at 25.	**S2**	Detect F3. Trigger interrupt. Deliver age-adjusted reframing. Resume S2.	**Y**
IT-05	4. Interrupt	**Belief+ (Clock)**	I check the clock 10+ times. At 3 AM I panic calculating hours left. Cannot stop.	**S3**	Detect monitoring. Trigger interrupt. Deliver clock-watching reframing + SCT.	**Y**

IT-07	4. Interrupt	**Belief+ (Safety)**	I cancelled gym and morning plans. I arrange life around possible bad nights.	**S3**	Detect safety behavior. Trigger interrupt. Deliver reframing. Resume S3.	**Y**
IT-08	4. Interrupt	**Belief+ (F1+F3)**	Only got 6.5h instead of 7. Should not go to presentation – I will embarrass myself.	**S2**	Detect multi-factor belief. Combined reframing. Resume S2.	**Y**
FP-01	5. False-Pos	**Normal tiredness**	I feel extremely exhausted today. I slept poorly and just want to get through the day.	**S2**	Normal complaint. NO interrupt. Empathetic acknowledgment. Continue plan.	**N**
FP-02	5. False-Pos	**Adaptive coping**	I slept badly but I have been through this. I will stick to routine. It will regulate.	**S3**	Adaptive cognition. NO interrupt. Reinforce positive mindset.	**N**
FP-04	5. False-Pos	**Health question**	Is lack of deep sleep linked to Alzheimer? I read something and got worried.	**S1**	Information-seeking. NO interrupt. Balanced factual response.	**N**
FP-08	5. False-Pos	**Grief**	I am sad today. My dog passed away last night. I was up with him at the vet.	**S3**	Acute grief. NO interrupt. Empathy. Offer to pause CBT-I plan.	**N**
PR-01	6. Recovery	**Clean resolution**	(After reframing) Okay, I understand. So what time should I go to bed tonight?	**Interrupt**	Detect resolution. Restore S2 from checkpoint. Resume prescription.	**N**
PR-03	6. Recovery	**User rejection**	No, you do not understand. You are just a chatbot. My situation is different.	**Interrupt**	Detect non-resolution. Validate. Return to data-driven approach. Restore S2.	**N**
PR-05	6. Recovery	**Double interrupt**	(During reframing) I will lose my job, my apartment... [catastrophic cascade]	**Interrupt**	Nested interrupt. Break down cascade step by step. Then resume original.	**Y**

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SECTION V : Proposal Information

Proposed Title *(Capitalise each word)* ***(Not more than 20 words)***

A Hybrid RAG-FSM Architecture for Internet-Delivered Cognitive Behavioral Therapy for Insomnia

Introduction *(Introduce the context & research gaps/problem)* ***(300-400 words)***

Insomnia disorder, also known as chronic insomnia, is a serious public health problem. International clinical guidelines recommend Cognitive Behavioral Therapy for Insomnia (CBT-I) as a first-line nonpharmacological intervention (Morin et al., 2006; Riemann et al., 2017). CBT-I contains a series of structured behavioral therapies, including sleep restriction, stimulus control, and sleep hygiene education, alongside cognitive restructuring targeting negative and dysfunctional sleep beliefs. A large amount of evidence shows that CBT-I can continuously improve sleep efficiency and insomnia severity (Ritterband et al., 2017).

Although the effect of face-to-face CBT-I has been confirmed, it is difficult to promote this therapy because there are not enough therapists, too many people waiting in line, the high cost of treatment and geographical restrictions (Espie et al., 2012). In response to this problem, internet-delivered CBT-I (iCBT-I) has been confirmed to be a viable alternative. It is scalable, effective and relatively low-cost (Ritterband et al., 2017; Zachariae et al., 2016). However, the current iCBT-I systems still face many obstacles, such as high patient attrition, poor treatment compliance, and a lack of strong natural language capabilities to provide personalized interaction (Koffel et al., 2016; Lin et al., 2025).

Large Language Models (LLMs) can generate empathetic dialogues and understand natural language, which bring new possibilities to the field of digital mental health, help to improve user participation, and can also provide personalized treatment content (Bao et al., 2025). However, directly deploying LLMs in clinical environment will also bring some serious risks, such as inaccurate suggestions, content hallucination and possible deviations from standard treatment process (Bedi et al., 2025; Mansoor, 2025). As for the existing LLM fine-tuning methods for CBT-I, they also face two main problems. The ability of clinical reasoning will decrease during long-term treatment, and the calculation expense is too high. (Chen et al., 2024).

One problem that this project wants to solve is how to balance the flexibility of LLM in dialogue with the clinical rigor of the structured treatment process and protocols. CBT-I is actually well suitable for implementing in a structured computation method, because its behavioral part runs step by step in the set time sequence according to clear rules. Yet the cognitive reconstruction in CBT-I requires a more flexible and situation-based response to those sudden dysfunctional ideas. This project proposes a hierarchical structure. This architecture combines LangGraph-based finite state machine (FSM) control to manage the behavioral stage, and also integrates retrieval enhancement

generation (RAG) technology to achieve clinically compliant cognitive intervention. This unified dialog framework can not only ensure that the standardized clinical protocols are strictly followed, but also respond in an adaptive manner.

Aims and Objectives *(What & how)* ***(100-150 words)***

The aim of this project is to design and develop an iCBT-I dialog system prototype, which integrates RAG technology and FSM workflow orchestration. The proposed architecture is expected to combine the flexibility of LLM and deterministic process control to support clinically compliant insomnia interventions. During the whole implementation of CBT-I, the system is able to improve the consistency of workflow, enhance the reliability of knowledge retrieval and protocol adherence.

Objectives:

- To develop an iCBT-I prototype, which integrates FSM workflow and RAG knowledge retrieval.
- To implement structured CBT-I intervention process based on the proposed architecture.
- To evaluate the prototype functionality through scenario-based testing and simulated dialogues.
- To validate the operating efficiency of the iCBT-I prototype using quantitative metrics.

Literature Review *(Related works)* ***(1000-1300 words)***

3.1. Clinical Foundations of CBT-I

CBT-I is a multicomponent psychological intervention method that integrates several behavioral strategies, including sleep restriction therapy (SRT), stimulation control therapy (SCT) and sleep hygiene education, as well as cognitive reconstruction techniques for those beliefs related to poor sleep (Morin et al., 2006). The American Academy of Sleep Medicine commissioned a comprehensive meta-analysis, which confirmed that CBT-I can stably improve many key sleep indicators, so it is recommended as an effective treatment for chronic insomnia (Morin et al., 2006). Epstein et al. (2012) further verified through a dismantling randomized controlled trial (RCT) that although a single behavioral component itself can also bring a certain improvement effect, combining multiple components for intervention can achieve the highest remission rate**,** and is desirable in clinical practice.

3.2 Evolution of Internet-Delivered CBT-I

The shift from face-to-face treatment to iCBT-I is promoted by step-by-step empirical validation. Ritterband et al. (2009) established the first confirmed the feasibility of fully automatic digital treatment that does not require the participation of therapists through the SHUTi project, a six-week web-based CBT-I project. Later, a large-scale RCT verified that this treatment effect continued in the follow-up visit after one year, which made SHUTi a benchmark for measuring the efficacy of iCBT-I (Ritterband et al., 2017). But at the same time, there are still problems of low treatment compliance and high patient dropout rate in digital treatment forms (Lin et al., 2025). Another mobile phone application named CBT-i Coach, which is a rule-based expert system, uses hard-coded

algorithms to calculate sleep prescriptions. A pilot trial of it showed that structured digital tools can well improve the retention rate of treatment (Koffel et al., 2016). However, as a purely deterministic system, CBT-i Coach lacks the ability to understand natural language and cannot carry out cognitive reconstruction. Because of this functional defect, people began to promote the integration of conversational artificial intelligence (AI) into the treatment of iCBT-I. Recently, Chiu et al. (2025) conducted an RCT to compare a CBT-I chatbot and a sleep education website to see how effective they are in psychiatric outpatients. The test results showed that the sleep quality and comprehensive psychological symptoms of the patients in the group of chatbots were highly improved. This also provided direct clinical evidence for chatbot-based CBT-I treatment, indicating its feasibility.

3.3 LLMs in Mental Health: Promise and Pitfalls

The emergence of Large Language Models (LLMs) has brought new possibilities to AI-driven mental health interventions, especially in the realization of natural language interaction and personalized dialogue generation. Inkster et al. (2018) provided some early real-world evidence that an empathy-driven conversational AI application called Wysa can well alleviate depressive symptoms, which also suggested that users are willing to accept text-based AI treatment support. Another example is Limbic Care, after large-scale RCT evaluation, it was found that compared with the form of static content, interactive CBT supported by LLM can greatly improve user participation, and personalized functions can also bring better symptom relief. (McFadyen et al., 2024). Gupta et al. (2024) showed that integrating Retrieval-Augmented Generation (RAG) into mental health chatbots can improve the reliability of high-risk intent classification, validating retrieval-based enhancement as a safety mechanism for clinical AI applications.

However, the critical limitations still exist. Bedi et al. (2025) systematically evaluated LLMs for CBT dialogue in both generation-only and RAG-enhanced settings. They found that while LLMs produce linguistically adequate therapeutic responses, they remain limited in maintaining treatment consistency and conveying empathy across sessions, indicating that the quality of language itself does not ensure the effectiveness of treatment. Chen et al. (2024) fine-tuned ChatGLM-6B with LoRA for a mobile CBT-I application, and reported encouraging initial outcomes, but they also pointed that parameter-efficient fine-tuning is prone to reasoning degradation in long-term multi-session treatment interactions. Most comprehensively, Bao et al. (2025) evaluated eight LLMs with three adaptive strategies on an RAG-enhanced iCBT-I dialogue system, and concluded that clinical reliability demands more than prompt engineering or model fine-tuning; additional architectural constraints are essential.

3.4 FSM and RAG: Architectural Solutions

Finite State Machines (FSMs) provide a well-established formalism for modelling systems with discrete states and deterministic transitions, making them well-suited for clinical workflow management where protocol compliance is essential. Recent work has begun to explore FSM-based

control within AI-driven therapeutic dialogue. Zhao et al. (2025) proposed FiSMiness, a paradigm that applies FSM to structure LLM-based emotional support conversations by defining states as the combination of user emotion and supporter strategy, demonstrating that FSM-guided generation outperforms both direct inference and chain-of-thought prompting on established benchmarks. Vijayan (2025) implemented a hybrid dialogue system, which integrates BiLSTM intent classification with FSM-controlled dialogue flow, specifically for structured health counselling. This work found that the FSM framework can be used in medical applications while taking into account both linguistic flexibility and operational reliability. In deployed clinical environments, the CLARITY triage system also confirmed that it is feasible to combine FSM dialogue management and LLM analysis on a large scale (CLARITY, 2025). LangGraph is a state machine framework based on graph structure, which is used to arrange LLM. It supports conditional routing, state checkpoints and interruptions of Human-Computer Interaction (HCI), which directly fit the requirements of CBT-I in behavioral stage management (LangChain, 2024).

Retrieval-Augmented Generation (RAG) method proposed by Lewis et al. (2020) solves a fundamental limitation of parametric LLM. RAG links the generated output with the retrieved external knowledge, which can reduce the subjective assumption of the model and improve the accuracy of the facts. In the field of mental health, RAG has been used to link LLM's answers with clinical guidelines and diagnostic standards in psychiatric diagnostic tasks (Mansoor, 2025). It can also support culturally adaptable CBT dialogue by retrieving information from the knowledge base (KB) of specific situations (Forane et al., 2025). Skill-RAG indicated that the model can use its own knowledge to filter out irrelevant retrieval contents, which is used to judge whether it can ensure that the cognitive intervention of CBT-I always maintains clinical applicability and situational relevance (Isoda, 2025).

Rather than separately deploying LangGraph or KBs, Lenh et al. (2024) showed that FSM and RAG can be integrated into the same mental health dialog system. In this system, a module controlled by FSM is responsible for managing depression assessment, and another RAG-based subsystem is responsible for handling open mental health queries, and the two can be flexibly switched through an intention detector. But this integration method is still limited to general depression support, and it does not solve several key problems required by CBT-I, such as structured time-dependence, deterministic sleep prescription calculation, or the global interruptible cognitive reconstruction.

Discussion and Summary

The rule-based systems can ensure clinical compliance of certainty, but they lack the ability to interact with natural language. The LLM-based system can improve the user's participation and dialogue flexibility, yet it cannot guarantee the strict adherence of the structured treatment plan. While the enhanced system based on RAG can improve the factual basis of the answer, the inherent time dependence in behavioral sleep intervention cannot be captured. Recent hybrid systems have

started integrating FSM with LLM or RAG components (Zhao et al., 2025; Lenh et al., 2024), but the problem is that these architectures just use flat state machines to manage relatively single tasks, such as questionnaires or emotional support. They cannot handle the multi-stage and time-dependent behavioral progress required in CBT-I. No current system has been able to combine hierarchical FSM-driven behavioral workflow orchestration with cognitive intervention, which is based on RAG separately tracking the behavioral layer and the cognitive layer in a framework specially designed for CBT-I. As shown in Figure 3.1, this project proposes a hierarchical RAG-FSM architecture. In this architecture, a multi-layered FSM is responsible for deterministic behavioral stage progression, while a cognition-focused RAG module runs as a global interrupt mechanism.

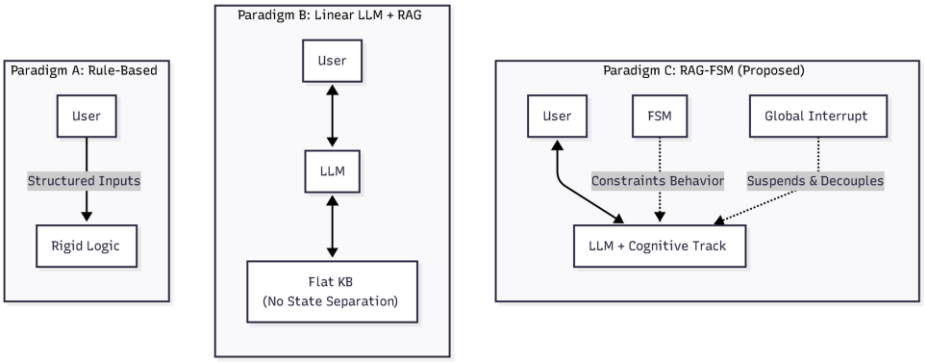


Figure 3.1 Conceptual comparison of rule-based, linear LLM-RAG, and the proposed FSM-RAG conversational architectures

Research Methodology

(Data collection / system design, tools & technologies / datasets, & Evaluation methods (Ex. testing, user study, simulation) ***(300-500 words)***

4.1 Prototype Development Method

This project uses the Design Science Research method. The whole process is to repeatedly build, demonstrate and evaluate the workpiece through an iterative cycle. An Agile SDLC method is used for system development to combine system implementation with iterative evaluation. Figure 4.1 shows the whole method framework and its corresponding stages.

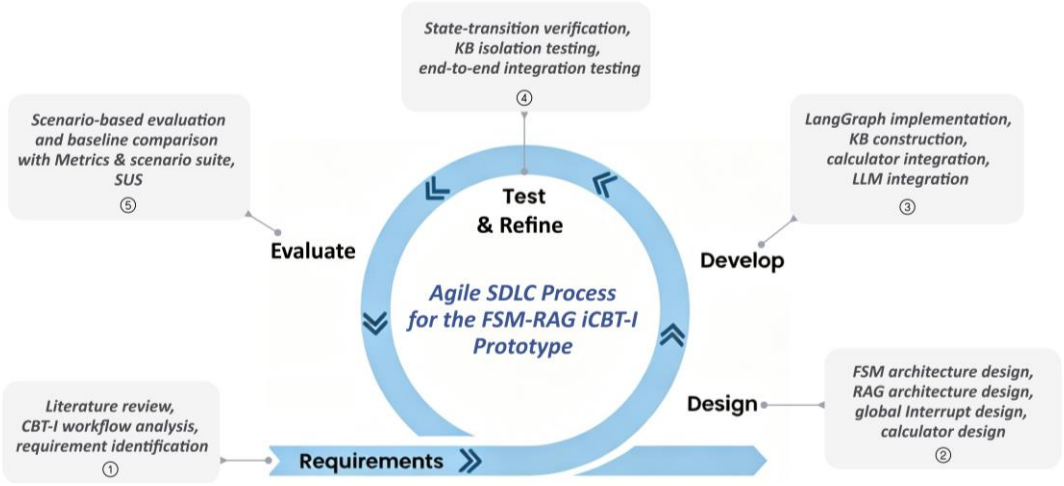


Figure 4.1 Stages of Agile SDLC and Corresponding Activities

4.2 System Architecture Design and Development

The prototype implements a hierarchical architecture containing four integrated components.

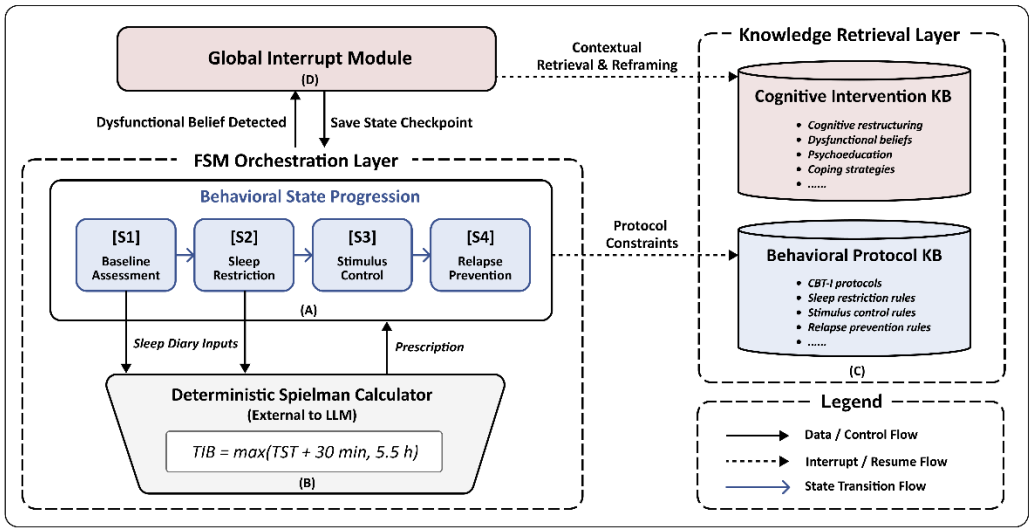


Figure 4.2 Hierarchical Architecture of the RAG-FSM iCBT-I System

A. The FSM Orchestration Layer uses LangGraph, which manages the four sequential execution and deterministic CBT-I stages ($\boldsymbol{S_1}$-$\boldsymbol{S_4}$) according to the established protocol.

B. In $\boldsymbol{S_1}$-$\boldsymbol{S_2}$, the system collects the sleep diary inputs through structured dialogue prompts, which then be sent to a deterministic Spielman calculator independent of LLM.

C. The Knowledge Retrieval Layer uses RAG, which is composed of two isolated vector KBs, Behavioral Protocol KB and Cognitive Intervention KB.

D. The Cognitive Interrupt Module detects the dysfunctional belief, it first suspends the currently running behavioral state, and call the cognitive intervention KB via LangGraph checkpoint, then resumes it back after processing.

4.4 Testing and Refinement

This stage need to iteratively verify the integrated system to ensure the system quality before the formal evaluation begins. The continuous testing work is divided into three parts:

(1) Check the state conversion during the execution process to validate the FSM logic.

(2) Perform boundary leakage testing to guarantee strict isolation between the dual KB.

(3) Conduct end-to-end integration testing to validate the fluid data flow, Spielman calculation synchronization, the RAG pipelines, and the LLM routing layer.

Prior to evaluation, the expected system behavior is specified for every test case, drawing from the CBT-I clinical protocol for FSM transition logic, the Spielman algorithm for prescription benchmarks, and published CBT-I therapist manuals for interrupt trigger.

4.5 Evaluation

(1) Architectural Metrics Selection

Table 4.5.1 presents four evaluation metrics, each mapped to one architectural component and one documented failure mode.

No.	Metric	Definition	Target	Architectural component	Failure mode addressed
(a)	FSM Transition Accuracy	Proportion of correct stage transitions	$\geq 95\%$	FSM Orchestration Layer	Conversational Protocol Drift (Bao et al., 2025)
(b)	Prescription Precision	Accuracy of the Spielman formula output	100%	Deterministic Calculator	LLM clinical arithmetic failure (Chen et al., 2024; Bao et al., 2025)
(c)	Interrupt Precision and Recall	Precision and recall of interrupt trigger decisions	$P \geq 80\%$, $R \geq 75\%$	Cognitive Interrupt Module	Unreliable trigger decisions (Gupta et al., 2024)
(d)	Contextual Recoverability	Correct recovery rate after each cognitive interrupt	$\geq 90\%$	FSM-Interrupt Interface	Inter-session context loss (Bedi et al., 2025)

Table 4.5.1 Evaluation Metrics and Validation Targets

(2) Scenario-Based Test Suite Preparation

Six scenario categories function as test environments targeting the four failure modes in Table 4.5.1. Scripted dialogues are applied in each category. Tests cover functional integration, state transition and retrieval-isolation integrity.

#	Category	Sample Case	Property Tested	Metric
1	Workflow progression	Compliant 7-day diary *"Bedtime: 11 PM; Wake-up: 6 AM; Awakenings: 0. SE is above 85% this week"*	FSM orchestration under compliant path	(a)
2	Workflow deviation	Off-topic tangents *"What is the weather today?"*	Resistance to protocol drift	(a)
3	Contradictory input	Logically conflicting values *"I woke up 15 times last night but my sleep efficiency was 98%."*	FSM verification and anomaly flagging	(a) (b)
4	Interrupt trigger	Clear dysfunctional statements *"If I don't fall asleep in 10 min, my whole week will be ruined."*	Cognitive interrupt triggering (Recall)	(c)
5	False-positive suppression	Standard complaints or negative mood text *"I feel extremely exhausted and irritable today due to poor sleep."*	Workflow continuity preservation (Precision)	(c)
6	Post-interrupt recovery	[Completed cognitive restructuring] *Let's return to your sleep window...*	Checkpoint-based context restoration	(d)

Table 4.5.2 Scenario Test Suit

(3) Baseline Simulation

A single-factor controlled experiment quantifies the FSM's architectural contribution. The independent variable is the presence or absence of LangGraph StateGraph routing; dependent variables are the four architectural metrics; control variables are prompt templates, model weights, and RAG content. The baseline runs the same LLM and RAG assets as a linear chain without FSM state management.

(4) Supplementary Usability Assessment

The usability assessment targets the software interface, not the therapeutic intervention. Peer usability testing recruits N = 5–10 volunteers via convenience sampling. Participants complete the System Usability Scale (SUS) after scripted sessions.

4.6 Tools, Technologies, and Data Sources

- LangGraph
- Python development environment
- Open-source LLM
- Vector database (e.g., FAISS)
- Scenario-based test scripts
- CBT-I clinical guidelines and intervention manuals
- System Usability Scale questionnaire

1

Scope & Limitations ***(100-150 words)***

The scope of this project is limited to the design and evaluation of an iCBT-I prototype based on FSM-RAG. This system only focuses on selected CBT-I modules containing sleep diary management, sleep restriction guidance, and cognitive restructuring support. The evaluation is for the performance metrics of the system itself and preliminary usability assessment, but its clinical effect or treatment results are not validated. As a prototype for research, it is not suitable for deployment in a formal clinical environment, and cannot replace professional medical services. A limited number of scripted scenarios are used in the evaluation process and a small convenience sample for usability testing, which will reduce the universality of the results a certain extent.

Potential Project Significance ***(100-200 words)***

From the theoretical view, the FSM-RAG architecture provides a model that can be reused. This model does not need to fine-tuning LLM, it can be integrated into the structured workflow in clinical practice, and meanwhile, it can ensure the safety of patients through the execution of certain constraints. It balances the reliability of rule-based systems and the flexibility of conversational AI, puts forward some design principles that can be migrated, and sets a scenario-based evaluation

framework for clinical AI architecture. From the practical view, this architecture can help build a stable and compliant iCBT-I system. The decision-making boundaries of this system are ethically transparent, which helps to overcome some limitations of existing online intervention tools. It improves digital insomnia intervention through local deployment models, so as to prevent sensitive health data from being transmitted to the outside, and this architecture can also be extended to other structured digital psychotherapy applications.

1

Expected Outcomes & Concluding Remarks

(Deliverables, expected results & conclusions) ***(200-300 words)***

7.1 Expected Outcomes

The tangible deliverable is a functional iCBT-I prototype application featuring:

1) A LangGraph-powered conversational interface that manages behavioral therapy progression through a Macro-FSM with checkpointed state persistence.
2) A dual-track RAG knowledge base serving both behavioral protocol retrieval and cognitive intervention retrieval.
3) An integrated sleep diary module with automated sleep prescription calculation following the clinically validated Spielman formula.
4) A cognitive interrupt mechanism that dynamically invokes targeted cognitive restructuring when dysfunctional sleep beliefs are detected.

Measurable outcomes and corresponding performance thresholds are listed in Table 4.2.1. Besides, a SUS score above 68 is targeted to indicate acceptable usability.

7.2 Conclusion

Solve the delivery problem of iCBT-I from the perspective of software architecture rather than simple model capability, this project contributes a novel and potentially generalizable design pattern for conversational AI that meets clinical standards. The hierarchical FSM-RAG architecture responds to the demand identified across the literature, that is, the need for an architectural constraint mechanism that can enable LLMs to operate safely in the evidence-based therapy framework. The evaluation strategy in the plan combines architecture performance metrics with clinical outcome measures, which can provide multidimensional evidence for the feasibility of the method. The successful demonstration of the prototype system is expected to provide a reference for the future supervised AI-based digital therapy design and support the responsible integration of generative AI into the mental health delivery systems.